UNIVERSITI TUNKU ABDUL RAHMAN

Faculty of Information and Communication Technology

UCCD3223 Mobile Applications Development

(February 2026 Trimester)

Group Assignment 1

Project Title: Grammar Expert

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Marking scheme	**Marks**	**Remarks**
Creativity/novelty of concept	**X 4**	
Completeness of idea	**X 3**	
Overall design	**X 2**	
Neat documentation layout	**X 1**	
TOTAL	**/100**	

1.0 Introduction

Nowadays, smartphones are very important in our daily lives. We use them to work, learn and talk with anyone. Therefore, mobile apps are the main reason for this change. They help use do many things easily from our phones. Now, many people like to use Artificial Intelligence (AI) because AI can understand and process language in a simple way such as ChatGpt and Google Gemini.

Our project will use AI to help everyone with their English writing. It is because writing clearly and correctly is very important. For example, students need to write essays or reports and so on. Besides that, workers also need it for their jobs. However, many people have big problems with English grammar. For this reason, many people are not native English speakers, so they need help in a simple way.

Many existing grammar apps such as Grammerly, Quilbot and others need to spend expensive monthly payments. It is difficult for everyone when they need help. Our app will be different. We want to make grammar help available to everyone for free.

Our proposed solution is to provide a mobile app called “ Grammar Expert”. It is a grammar helper app. This mobile app will use AI to find out the mistakes and suggest better ways to correct the mistakes. It also will provide explanations so users can understand and learn from their mistakes. This action will let users go-ahead in their English grammar level.

This app supports two Sustainable Development Goals (SDGs) . The first goal is quality education. It is because our app provides an easy way for the people to improve their English skills anytime and anywhere in their daily lives. The second goal is to reduce inequalities. Our app is free for use. Therefore, it can give them a better chance to succeed in school and work and also help them to close the gap between rich and poor.

2.1 Problem Statement

For the last ten years, mobile phones have changed how we live, work, and talk to each other. However, even if technology has advanced better, mobile users still have trouble with linguistic accuracy. While Artificial Intelligence (AI) has brought in a new wave of innovation, many writing tools available today still function as "black boxes" that fail to provide the level of intelligence and personalization that students actually need. When a user makes a mistake, these programs generally fix it automatically without telling them why. Users can't learn from their mistakes if they don't get educational feedback. This makes them rely on software instead of improving their own reading skills, which is not what Quality Education is all about.

Furthermore, current grammar tools aren't really made for a mobile-first experience. A number of them are basically desktop apps that have been cut down so they don't operate well on small displays. Next, these apps usually don't have context-aware intelligence because they can't tell when to use a formal tone for schoolwork and when to use a relaxed tone for personal conversation. Without an application that bridges the gap between simple error correction and educational guidance, students will remain limited in their ability to communicate effectively.

2.2 Motivation

The primary motivation for developing the Grammar Helper App (Grammar Expert) is to explore the next wave of innovation through the rise of Artificial Intelligence. It's especially interesting to us how new AI technologies, like OpenAI's ChatGPT and Google's Gemini, can give mobile apps a level of intelligence and customization that wasn't possible with older software. We want to show that these high-tech systems can run smart agents that can do more than just work as tools. These tools enable us to transform a basic utility app into a powerful learning platform. We want users to understand "why" they made mistakes in their work instead of just having them corrected.

We are also motivated by the idea that innovation can take many forms and does not always need to be groundbreaking to be valuable. We believe that true value is found in finding a specific target and positioning an application to properly address their needs. We want to make grammar

checking on mobile phones more educational and easy to use by choosing to refine or improve solutions that are already out there.

2.3 Market Gap Analysis

While there are many different writing tools available today, there are still many products that do not meet the specific needs of every user. We can see a big gap in the market where students and learners aren't getting served properly. Most grammar apps that are already out there focus on being well-known and making money, which usually means they put "quick fixes" and automated corrections over actual learning. This leaves a group of users, mostly students who need a tool that works more like a personal tutor than just a simple spell checker. The market is currently saturated with tools that fix errors without explaining them, which does nothing to help the user grow or improve their independent writing skills.

Furthermore, many advanced AI tools that are capable of providing deep analysis are either too expensive or restricted to desktop browsers, making them inaccessible for many students on the go. There is a clear lack of mobile-first applications that use advanced chatbot systems to provide educational feedback. Our app fills this gap by offering a mobile-friendly experience that focuses on helping users learn rather than simply fixing things. By focusing on contextual intelligence and the "why" behind every grammatical rule, we provide alternative generic apps that dominate the current market, directly serving users who are committed to improving their skills.

3.1 Proposed Ideas

Based on the problem identification above, there are several issues addressed, and this project aims to have the development of a mobile application with integration of an AI-Powered education grammar tutor with mobile first concept due to the increased spending time of people on social media. This app will act as an interactive AI writing tutor that can explain grammar mistakes, provide real-time feedback, as well as helps user to improve writing skills compared to traditional grammar helpers that act as a silent auto-correct tool. These features will be created with consideration of the interactive characteristics of mobile applications.

Firstly, a real-time grammar highlighting feature that will be developed in this grammar helper mobile app can check grammatical errors in English for the user, and highlight the words, phrases or sentences where errors are found. With the highlighted grammatical errors shown on screen, this will navigate the user to click on the highlighted area, and the hidden note of error explanation and ways to fix the error will appear beside the highlighted area. For example, when the user enters "How does you go to school?", then the system will automatically highlight the "does" in the sentence and mention why it is wrong with the correction. For example, it should replace "does" with "do". The highlight will disappear when errors are corrected with some explanation and examples. This functionality needs to be included in the grammar helper because it provides users with a clear explanation of grammar, such as tenses, which can help them improve their English.

Secondly, an AI explanation chatbot will also be developed in the app, which allows users to ask grammar questions and receive clear clarification to clear their doubts. The user utilized these features to help them build a deep concept and the basics of English. The user can enter a prompt at the chatbot with clear requirements and instructions on what they do not understand, which can create a better understanding of the concept while they are learning. For instance, questions such as "Why is it wrong?" after a correction session, and the chatbot will respond to them in beginner-friendly ways. With this AI tutor integrated into this app, the user can easily improve their English grammar in a more flexible way with just a smartphone, anytime and anywhere.

Lastly, the app provides customization to the user with context-aware writing modes that allow AI to adapt corrections based on the writing purpose. The user can use this function to ensure

their grammar used for a certain situation is suitable. This app not only allows users to check grammar, but also allows the user to select a suitable mode from multiple writing contexts, such as academic mode, professional mode, casual mode and email mode. For example, a lecturer who want to do research and check for grammar, then the academic mode under this function can be considered as an option as the formality can be changed by the word without changing the content.

Provide a visualized dashboard for the user for the descriptive summary on how much time they have spent on learning and the progress and improvement they have made. This dashboard allows them to check their progress in learning English and also record the error pattern they always made into a note session. This function is important as it can help users to check up their progress and identify similar errors that they always made, which can help them to addresses on these mistakes that always made.

3.2 Solutions

3.2.1 Enhancing Learning Through Personalized Error Pattern Detection

The applications with the implementation of error pattern detection and personalized learning recommendations provide users with a more flexible learning environment, rather than just providing the correction of grammar with simple explanations, which could create dependency for the users. Similar types of grammatical errors, such as spelling, verb tense inconsistency, article misuse, or incorrect preposition selection, are repeatedly made by users. However, existing applications cannot really identify the recurring error patterns due to a lack of linkage between the cases and errors.

The proposed system will track user error frequency and categorize common mistake types. Then, the application will generate targeted short learning modules within two to ten minutes lesson designed to address specific weaknesses based on the analysis. For instance, the user would be recommended a short lesson with precise exercises on definite and indefinite article usage if articles frequently misused by user. This personalized learning strategy can promote long-term improvement by transforming recurring mistakes into structured learning

opportunities. User experiences would be improved by providing interaction with users, encouraging independent skills development and promoting cognitive engagement with a shift from reactive correction to proactive educational support instead of AI automation dependence.

3.2.2 Designing a Mobile-First Real-Time Feedback System

This application will develop a mobile-first design philosophy that will be designed specifically for smartphones' lightweight characteristics, focusing on interaction, speed, clarity, and usability to overcome the limitations of the existing approaches. Compared to the existing application design, usually a simplified version on desktop, this mobile app design provides real-time feedback by applying real-time, lightweight grammar highlighting during typing. Moreover, tap-to-expand explanation features are also included in compact pop-up panels for different situations. To improve the user experience, this application will have a split-screen layout combining writing space and AI feedback that allows users to received a quick feedback or suggestions, which can be easily viewed and corrected. An efficient API integration is also included to ensure minimal processing delay after the user sends a message to the system. Thus, this approach enhances user experience while maintaining advanced AI-driven capabilities, which ensure smooth performance and accessibility compared to desktop functionality.

3.2.3 Incorporating Context-Aware Intelligence

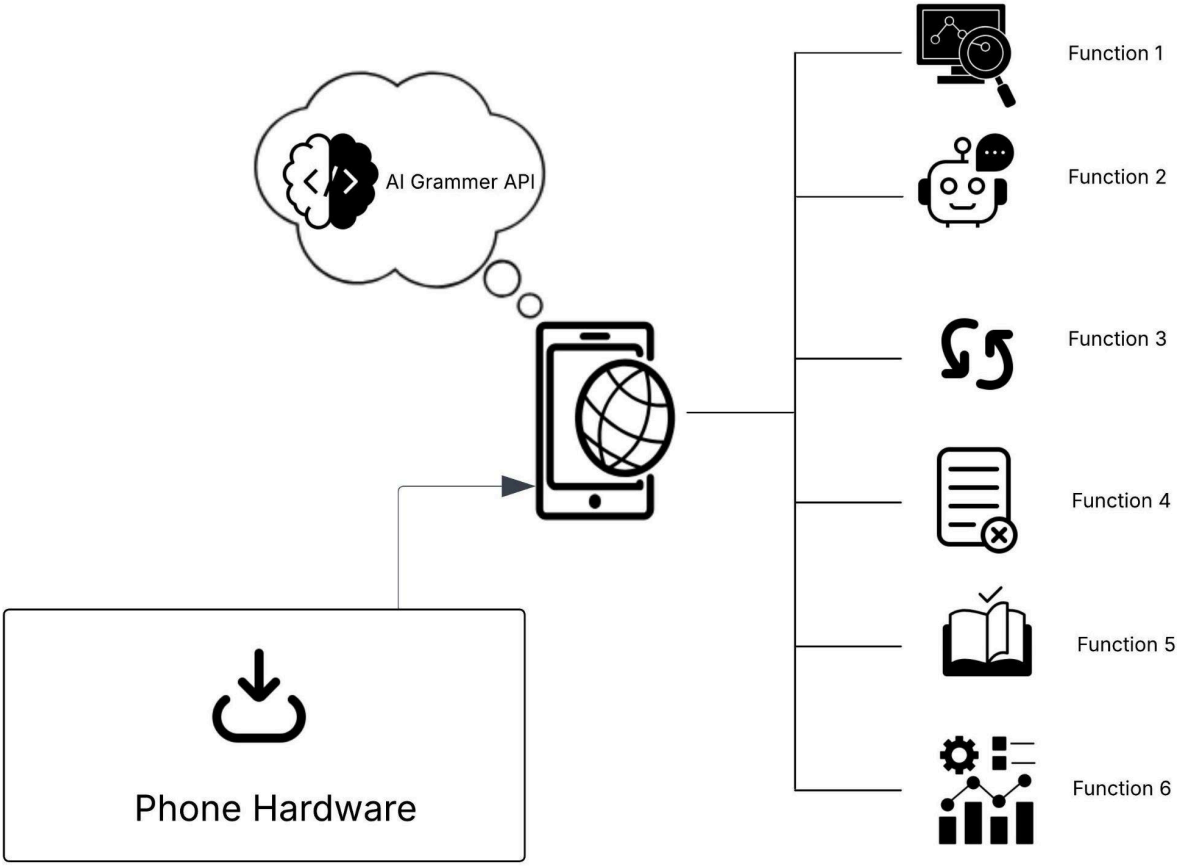
The application integrates context-aware writing modes, allowing users to select writing purposes such as academic, professional, email, or casual communication to improve on the existing application. This is because existing English grammar tools only focus on structural accuracy without considering the communicative context of the writing, which will cause inappropriateness for different situations. With the considerations on the communicative context, the AI system will automatically adjust recommendations accordingly, modifying vocabulary suggestions, sentence complexity, and tone refinement based on the selected context from users, which is more scalable. For instance, informal expressions may be reformulated in academic mode to ensure appropriate professional communication. This functionality expands the scope of grammar checking from structural correction to communicative competence development while strengthening ethical and situational awareness in written communication.

3.3 Contributions

In conclusion, this app can benefit in four main areas, which include educational innovation, technological advancement, user experience improvement and market differentiation. Firstly, this application allows users to transform grammar checking into an explainable, interactive learning experience that aligns with lifelong learning principles, compared to existing applications that provide short-term error elimination. Secondly, by integrating AI-powered analysis for error detection, personalization for grammar practice, and mobile-first design into a lightweight and responsive application, which provides convenience to the user to have personalized learning experience if the user logs in with their mobile phone. The combination of external AI APIs and local data storage illustrates the practical implementation of intelligent agent systems in mobile computing, which will bring advancement in the technological area. Next, this application will also provide a better user experience by delivering real-time feedback and intuitive interaction to the users. Additionally, a progress visualization also included, which is specifically designed for smartphone usage. Lastly, this application addressing gaps left by existing grammar tools by focusing on context awareness, personalization, and educational depth, which is especially suitable for university students and non-native English speakers who seek long-term writing improvement. By combine artificial intelligence and educational methodology within a mobile environment, this project is better that common conventional grammar correction, as it introduces a more useful, thoughtful, and user-centered writing assistant. It has improved the function of existing grammar applications from passive editing tools to active learning platforms that support long-term communication effectively.

4. Initial Design of Solution

4.1 Overall System Architecture



Block Diagram

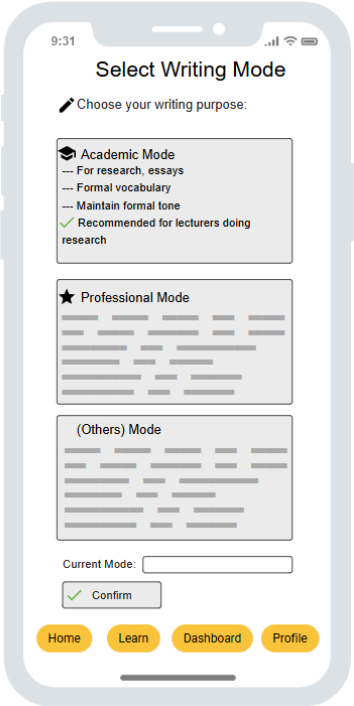
Component	Description
AI Grammar API	Provides grammar checking with Internet
Function 1 : Real-Time Grammar Highlighting	Automatically detects and highlights grammar errors as the user type and provide explanation
Function 2: AI Explanation Chatbot	Users can ask grammar questions and receive simple explanation
Function 3: Context-Aware Writing Modes	Users can select the writing modes then AI will

	adapt corrections and suggestions accordingly
Function 4: Error Pattern Detection	Tracks all user grammar checks over time, analyzes and identifies recurring mistake patterns
Function 5: Personalized Learning Modules	Short, target lessons automatically generated based on their error mistakes
Function 6: Visualized Dashboard	Display user's learning progress, time spent, error patterns and improvement over time
Phone Hardware : Local Storage	Stores history, settings and save checks

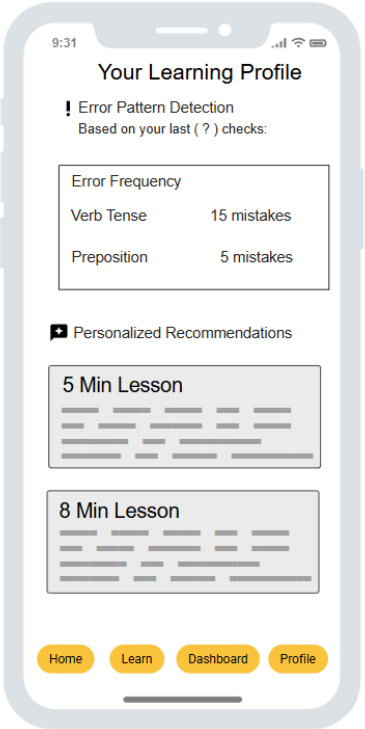
4.2 Mock-up UI Design

Screen 1: Home Screen with Grammar Highlighting

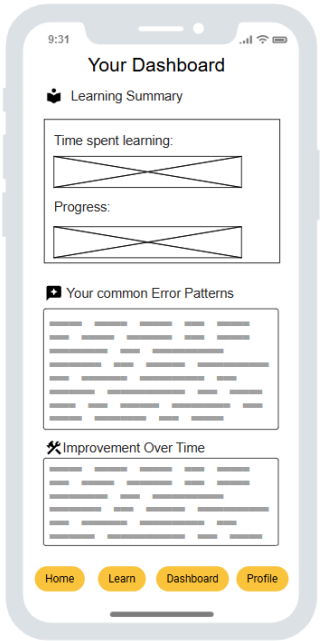
Screen 2: AI Explanation Real- Time Chatbot



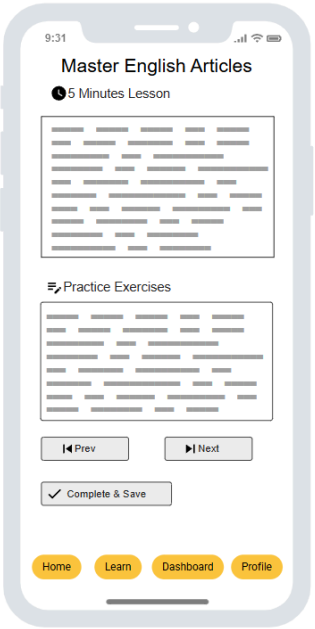
Screen 3: Context-Aware Writing Modes



Screen 4 : Error Detection & Personalized Learning



Screen 5 : Visualized Dashboard



Screen 6: Mini-Lesson Screen